

SaaS Customer Churn & Revenue Impact Analysis

End-to-End SQL Data Analytics Project

Executive Summary

This report presents a comprehensive analysis of customer churn and revenue impact for a SaaS company using transactional data from January 2024 to January 2025. The analysis was conducted using PostgreSQL across three relational tables: customers, subscriptions, and payments (80 customers total).

Business Problem

A SaaS company has experienced an increasing number of customers cancelling their subscriptions (customer churn). This situation threatens Monthly Recurring Revenue (MRR), increases customer acquisition costs, and slows business growth.

Management requires data-driven insights to understand churn patterns, identify high-risk customer segments, and estimate the financial impact of customer churn.

Business Objectives

The objectives of this analysis are:

- Analyze monthly revenue trends.
- Measure customer churn rate.
- Identify customer segments with the highest churn.
- Estimate recurring revenue loss.
- Measure Average Revenue Per User (ARPU).
- Provide actionable business recommendations.

Dataset Overview

Dataset Description

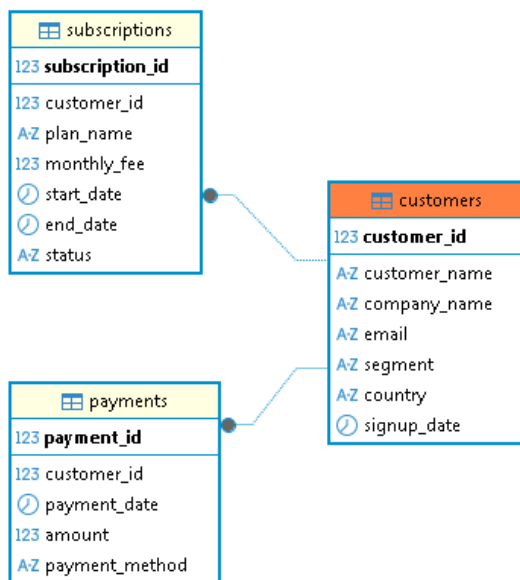
The analysis uses three relational tables.

Table	Description
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Customers	Customer profile information
Subscriptions	Subscription status and monthly fees
Payments	Customer payment history

Item	Information
Database	PostgreSQL
Number of Tables	3
Primary Key	customer_id

Data Understanding



The dataset consists of three relational tables: **customers**, **subscriptions**, and **payments**. These tables are connected through the **customer_id** field, enabling customer profiles, subscription information, and payment transactions to be analyzed together.

The database structure supports customer retention analysis by combining customer demographics, subscription status, and revenue-related information.

Data Analysis

Monthly Revenue Trend

Business Question

What is the monthly revenue trend?

Business Purpose

Analyzing monthly revenue trends helps management evaluate business performance over time, monitor revenue growth, and identify potential declines that could affect business sustainability.

SQL Query

```
select to_char(date_trunc('month', payment_date), 'mon yyyy') as month,
sum(amount) as Revenue from payments
group by date_trunc('month', payment_date)
order by date_trunc('month', payment_date);
```

Query Result

	A-Z month	123 revenue		A-Z month	123 revenue
1	jan 2024	2,396	7	jul 2024	30,930
2	feb 2024	6,186	8	aug 2024	29,432
3	mar 2024	12,674	9	sep 2024	23,847
4	apr 2024	16,563	10	oct 2024	19,357
5	may 2024	23,348	11	nov 2024	12,471
6	jun 2024	27,939	12	dec 2024	8,582
7	jul 2024	30,930	13	jan 2025	1,795
8	aug 2024	29,432			

Customer Churn Rate

Business Question

What is the customer churn rate?

Business Purpose

Measuring the customer churn rate helps management understand the percentage of customers who have canceled their subscriptions. This metric is essential for evaluating customer retention performance and identifying potential risks to recurring revenue.

SQL Query

```
select count(case when status = 'churned' then 1 end) as churned_customers,
count(*) as total_customers,
round (count(case when status = 'churned' then 1 end) * 100.00 / count (*), 2 ) as
Churn_rate from subscriptions;
```

Query Result

	123 churned_customers	123 total_customers	123 churn_rate
1	16	80	20

Business Insight

The analysis shows that the company has **16 churned customers out of 80 total customers**, resulting in a **20.00% churn rate**.

Business Interpretation

A churn rate of **20%** indicates that **one out of every five customers** has discontinued their subscription. This level of customer loss requires management attention because it can directly reduce recurring revenue and increase customer acquisition costs needed to replace lost customers.

Highest Churn Customer Segment

Business Question

Which customer segment has the highest churn?

Business Purpose

Identifying the customer segment with the highest churn helps management understand which groups are most likely to cancel their subscriptions. These insights support the development of targeted customer retention strategies.

SQL Query

```
select c. segment, count(*) as churn_count from customers c join subscriptions s on
c. customer_id = s. customer_id
where status = 'churned' group by c. segment order by churn_count desc;
```

Query Result

	AZ segment	123 churn_count
1	Enterprise	7
2	SMB	7
3	Startup	2

Business Insight

The analysis indicates that the **Enterprise** and **SMB** customer segments have the highest number of churned customers compared to other segments.

Business Interpretation

The high churn within the Enterprise and SMB segments suggests that these customer groups are at greater risk of leaving the service. Management should investigate the underlying causes of churn and implement targeted retention initiatives to improve customer loyalty within these segments.

Revenue Lost Due to Churn

Business Question

How much revenue is lost due to customer churn?

Business Purpose

Estimating revenue lost due to churn helps management quantify the financial impact of customer cancellations and understand how churn affects recurring revenue.

SQL Query

```
select sum(monthly_fee) as lost_revenue from subscriptions where status = 'churned';
```

Query Result

	123 lost_revenue
1	9,284

Business Insight

The analysis shows that churned customers represent an estimated **monthly recurring revenue loss of \$9,284**.

Business Interpretation

The loss of **\$9,284 in monthly recurring revenue** demonstrates the direct financial impact of customer churn. Reducing churn would help protect recurring revenue, improve business stability, and lower the cost of acquiring replacement customers.

Average Revenue Per User (ARPU)

Business Question

What is the Average Revenue Per User (ARPU) by customer segment?

Business Purpose

Analyzing Average Revenue Per User (ARPU) by customer segment helps management identify which customer groups contribute the highest average revenue and supports strategic decisions for customer acquisition and retention.

SQL Query

```
select c. segment, round(avg(p.amount),2) as ARPU from customers c join payments p
on c.customer_id = p.customer_id
group by c.segment
order by ARPU DESC;
```

Query Result

	AZ segment	123 arpu	
1	Enterprise	999	
2	SMB	299	
3	Startup	99	

Business Insight

The analysis shows that the **Enterprise** customer segment generates the highest **Average Revenue Per User (ARPU)** compared to the other customer segments.

Business Interpretation

The higher ARPU of the Enterprise segment indicates that these customers provide greater business value than other segments. Prioritizing customer retention and expanding relationships within this segment can significantly increase long-term revenue and profitability

Analysis Summary

Analysis	Result	Business Impact
Monthly Revenue Trend	Growth (Jan-Aug), then decline (Aug-Jan)	Mixed - Early Growth, Later Risk
Customer Churn Rate	20% (16 of 80 customers)	Revenue Risk
Highest Churn Customer Segment	Enterprise & SMB	High Retention Priority
Revenue Lost Due to Churn	\$9,284 Monthly Recurring Revenue	Financial Impact
Highest ARPU by Segment	Enterprise	High-Value Customers

Key Findings

Based on the analysis of customer, subscription, and payment data, the following key findings were identified:

1. Monthly revenue showed a positive trend from January through August, but declined from August into January of the following year.
2. The company's **customer churn rate is 20%**, meaning that **16 out of 80 customers** have canceled their subscriptions.
3. The **Enterprise** and **SMB** segments have the highest number of churned customers compared to the other customer segments.
4. Churned customers result in an estimated **Monthly Recurring Revenue (MRR) loss of \$9,284**.
5. The **Enterprise** segment generates the highest **Average Revenue Per User (ARPU)**, making it the largest contributor to the company's revenue among all customer segments.

Business Recommendations

1. Focus Retention Strategies on the Enterprise and SMB Segments

Since these two customer segments experience the highest churn, the company should evaluate their specific needs and strengthen customer retention programs to reduce the risk of customer loss.

2. Analyze the Root Causes of Customer Churn

The company should collect feedback from churned customers through surveys or interviews to identify the primary reasons behind subscription cancellations.

3. Monitor the Customer Churn Rate Regularly

A churn rate of **20%** represents a significant risk to recurring revenue growth. Therefore, customer churn should be established as a key performance indicator (KPI) and monitored on a monthly basis.

4. Prioritize High-Value Customers

Because the Enterprise segment generates the highest ARPU, the company should prioritize retaining these customers by improving service quality, expanding customer success initiatives, and providing a more personalized customer experience.

5. Reduce Revenue Loss Caused by Customer Churn

With an estimated monthly recurring revenue loss of **\$9,284**, the company should implement proactive churn prevention strategies to protect recurring revenue and support long-term business growth.